

Appendix A

Rules of Summation

A.1 SUBSCRIPTS

In the discussion to follow, let X and Y stand for numerical variables. Ordinarily, the numerical values for these variables are scores in some given set of data, in which each individual observed in a group of N cases is paired either with a single score, such as x , or with a pair of scores (x, y) .

The *subscript* notation is useful in keeping track of the various score values across the different individuals or observations. For the moment, we will consider N distinct individuals, arbitrarily indexed by the set of numbers $(1, \dots, i, \dots, N)$. The "running subscript" i denotes any one of these individuals. Then, X_i denotes the particular value of X associated with any particular individual i . That is, the variable X_i ranges only over the set of particular values

$$(x_1, x_2, \dots, x_N).$$

Now we need a way to discuss the *sum* of these particular values of X over the entire set of N different individuals i . The symbol $\sum$ (capital Greek sigma) stands for the operation "the sum of," referring to the values symbolized immediately after the summation sign. Thus,

$$\sum_{i=1}^N x_i$$

is read as “the sum of the N values x_i , beginning with $i = 1$ and ending with $i = N$.” That is,

$$\sum_{i=1}^N x_i = x_1 + x_2 + \cdots + x_N.$$

Suppose, for example, that $N = 4$, and the actual values associated with the four different individuals were

6, 8, 3, 11.

Then,

$$\sum_{i=1}^N x_i = 6 + 8 + 3 + 11 = 28.$$

In many simple contexts in which it is clear that the sum extends across *all* of a set of N individuals, the subscripts and superscripts on the summation sign are simplified or omitted. Thus, we often write:

$$\sum_i x_i$$

to indicate the sum of values across all N individuals i , or even

$$\sum x_i$$

to stand for this sum. However, the subscript is usually omitted only in very simple expressions where the set of values to be included in the sum is perfectly clear.

Quite frequently, symbols for values corresponding to individual observations are given two (or more) subscripts. That is, it happens that sometimes we want to discuss individual observations arranged into several groupings or cross-classifications. Then two (or more) sets of index numbers are used. For example, suppose that there were some J different *groups* of observations; these different groups are identified by being paired arbitrarily with the numbers

$(1, \dots, j, \dots, J)$.

Furthermore, within any particular group j , there is some number n_j of distinct individuals or observations, each of which is associated with an X score. Then, the particular value of the score for the i th individual in the group j is symbolized by x_{ij} . If we wanted to symbolize the sum of the scores for the particular group j , then we would write

$$\sum_{i=1}^{n_j} x_{ij}$$

which stands for “the sum of values for all of the n_j different individuals in the particular group labeled j .” Where it is clear that all individual scores in a particular group j are to be summed, this can be abbreviated to

$$\sum_i x_{ij}.$$

Notice that this quite literally is the symbol for "sum over i " for a set of scores x_{ij} ; the fact that j appears as a subscript for x but that i alone appears below the sigma is a cue that all the scores summed in this particular operation *must* have the same subscript j (that is, must be in the same group of observations).

However, often one wants to indicate the sum of X_{ij} over *all* individuals in *all* groups. Written out in full, this would be symbolized by

$$\sum_{j=1}^J \sum_{i=1}^{n_j} x_{ij} = (x_{11} + x_{21} + \cdots + x_{n_{j-1},j} + x_{n_j,j}).$$

This *double* summation is interpreted as "Take a particular group j , and sum scores over all the n_j different observations in that group; then, having done this for each of the J different groups, sum the results over all of these groups." Once again, this can be simplified to

$$\sum_j \sum_i x_{ij}$$

when the meaning of the symbol is perfectly clear in terms of which values are to be summed.

This idea can be extended, although in our discussions a *triple* summation is the largest such symbol used. In some problems, each individual i belongs simultaneously to *two* groupings. That is, suppose that all the scores in a set of data were arranged into a table with K rows, J columns, and JK cells, and that each individual observation belongs to one and only one row, one and only one column, and one and only one cell in the data table. Now let the various *column* groupings be symbolized by the numbers

$$(1, \cdots, j, \cdots, J).$$

The *rows* of the data table are indexed by the set of numbers

$$(1, \cdots, k, \cdots, K).$$

Then any cell of the table is indexed by a *pair* of numbers jk . The score of the i th individual in the cell jk of the data table is thus x_{ijk} . The number of individuals in the cell jk is shown by n_{jk} .

Now, when there are three subscripts like this,

$$\sum_{j=1}^J \sum_{k=1}^K \sum_{i=1}^{n_{jk}} x_{ijk},$$

the following is symbolized: "Take a particular cell jk and sum the scores of all individuals i in that cell; then, still within a particular column j , sum the results of doing this for each row k over the K respective rows; finally, sum the results for each column j over the J respective columns." A little thought should convince you that this is simply the sum of all individual scores in all cells, or the sum of all scores in the table.

In this instance, it is immaterial whether we write the sum for the rows or the sum for the columns first, because the total number symbolized is the same. However, this is not universally true, and the order of the summation signs can be important.

Sometimes we want to symbolize only the sum for a particular cell, row, or column: First of all, letting n stand for the number in cell jk , we write

$$\sum_{i=1}^n x_{ijk} = (x_{1jk} + \cdots + x_{njk}).$$

This denotes the sum of all individual scores in the *single* cell jk . If we write

$$\sum_{j=1}^J \sum_{i=1}^n x_{ijk} = (x_{11k} + x_{21k} + \cdots + x_{njk}),$$

we intend that the first the sum of individuals in a particular cell in column j and row k shall be taken, and then, still within the particular row k , the result for all the different columns j shall be summed. This amounts to the sum of all scores within the particular row k . On the other hand, if we want the sum of all the scores in a particular column j , we could write

$$\sum_{k=1}^K \sum_{j=1}^n x_{ijk} = (x_{1j1} + \cdots + x_{njk})$$

indicating the sum for a particular cell jk , and then the sum over the results for all rows k within the particular column j . In short, unless a subscript appears both in the symbol for the x values and under the summation sign, the total value symbolized pertains to some particular set of observations, represented by that subscript *not appearing* under the summation symbol. Notice that when the sum of all observations in a particular row k of the data table was intended, the sum was indicated over i and over j , but *not* over k . On the other hand, when the sum of observations in some column j was desired, the sum was indicated over i and over k , but not over j .

In learning to work with summation signs, learn the habit of reading the summation symbols from *right to left*, and from *inside the punctuation outward*. The *innermost* summation indicated in a mathematical expression is ordinarily to be carried out first, then the next summation symbolized to the left, and then finally the summation appearing closest to the margin on the left. Furthermore, although we have symbolized a row by k and a column by j in this discussion, this symbolism is arbitrary and, often, we will use j for rows and k for columns.

A.2 RULES OF SUMMATION

A few simple rules can help one learn to interpret and carry out the computation symbolized by statistical expressions involving summations. Furthermore, elementary statistical derivations almost always involve one or more of these summation rules used as an algebraic equivalence. First of all, some simple rules are given that apply when values for only one variable are being summed, and then we will consider situations involving two or more variables.

RULE 1. If a is some constant value over the N different observations i , then

$$\sum_{i=1}^N a = Na.$$

For example, suppose that 5 observations are considered, and paired with each observation is the constant number 10. Then

$$\sum_i a = 10 + 10 + 10 + 10 + 10 = 5 \times 10.$$

Anytime that, in the particular sum indicated, each and every individual observation entering into the sum is paired with exactly the same constant value a , then the sum is equal to the number of individuals considered times the value of a .

Extending this rule to double or triple summations, we have

$$\sum_{j=1}^J \sum_{i=1}^{n_j} a = \sum_{j=1}^J (n_j)a$$

$$\sum_{k=1}^K \sum_{j=1}^J \sum_{i=1}^{n_{jk}} a = \sum_{k=1}^K \sum_{j=1}^J (n_{jk})a$$

and so on.

RULE 2. Given the value a , which is constant over all individuals entering into the summation, then

$$\sum_{i=1}^N ax_i = a \sum_{i=1}^N x_i.$$

For example, consider the five individual scores
6, 7, 3, 9, 20.

Suppose that each of these scores is multiplied by the constant 2. Then

$$\begin{aligned} \sum_i 2x_i &= 2(6) + 2(7) + 2(3) + 2(9) + 2(20) \\ &= 2(6 + 7 + 3 + 9 + 20) = 2 \sum_i x_i. \end{aligned}$$

Once again, this rule can be applied to several sums:

$$\sum_{j=1}^J \sum_{i=1}^{n_j} ax_{ij} = a \sum_{i=1}^J \sum_{i=1}^{n_j} x_{ij}.$$

Notice here that the a symbol has no subscript, and thus may be regarded as a constant over the sum. However, it might happen that the sum is such that a_j appears instead of

a , meaning that a_j is a constant for all observations i in the *same* group j , but is different for different groups. Then,

$$\sum_{j=1}^J \sum_{i=1}^{n_j} a_j x_{ij} = \sum_{j=1}^J a_j \left(\sum_{i=1}^{n_j} x_{ij} \right).$$

One must be sure to notice the set of observations for which some value is constant in multiple summations.

RULE 3. If some operation is to be carried out on the individual values of X before the summation, this is indicated by mathematical punctuation, such as parentheses, or equivalent symbols having the force of punctuation. Unless the summation symbol is included within this punctuation, the summation is to be done after the other operation: for example,

$$\sum_i^N (x_i)^2 = \sum_i^N x_i^2 = (x_1^2 + x_2^2 + \cdots + x_N^2)$$

$$\sum_i \sqrt{x_i} = \sqrt{x_1} + \sqrt{x_2} + \cdots + \sqrt{x_N}.$$

However,

$$\left(\sum_i x_i \right)^2 = (x_1 + \cdots + x_N)^2$$

$$\sqrt{\sum_i x_i} = \sqrt{x_1 + \cdots + x_N}.$$

Be sure to notice that where the parentheses (or other symbols such as exponent and radical signs having the same force as parentheses) happen to appear makes a big difference in the sum symbolized. In general,

$$\sum_i x_i^2 \neq \left(\sum_i x_i \right)^2$$

$$\sum_i \sqrt{x_i} \neq \sqrt{\sum_i x_i}$$

$$\log \left(\sum_i x_i \right) \neq \sum_i \log x_i,$$

and so on.

This is especially important with multiple summation. For example,

$$\sum_{j=1} \left(\sum_i^n x_{ij} \right)^2 = \sum_j (x_{1j} + \cdots + x_{nj})^2.$$

The notation here tells us *first* to take the sum of all the individual values in a single group j ; then, having done this for each group j , we square the *sum* for each group; finally, the sum of the *squared* sums for the various groups is found. Squaring at the wrong place in this sequence of steps yields quite a different result.

RULE 4. If the only operation to be carried out before a sum is taken is itself a sum (or difference), then the summation may be distributed. Thus,

$$\sum_i (x_i - 4) = \sum_i x_i - \sum_i 4$$

$$\sum_i (x_i^2 + 3x_i + 10) = \sum_i x_i^2 + \sum_i 3x_i + \sum_i 10.$$

This rule is very important in algebraic manipulations on expressions involving sums. For example, using this rule we can make the following algebraic changes:

$$\begin{aligned} \sum_{j=1}^J \sum_{i=1}^{n_j} (x_{ij} - m)^2 &= \sum_{j=1}^J \sum_{i=1}^{n_j} (x_{ij}^2 - 2mx_{ij} + m^2) \\ &= \sum_{j=1}^J \sum_{i=1}^{n_j} x_{ij}^2 - \sum_{j=1}^J \sum_{i=1}^{n_j} 2mx_{ij} + \sum_{j=1}^J \sum_{i=1}^{n_j} m^2. \end{aligned}$$

Invoking Rules 1 and 2, we can reduce this expression to

$$\sum_j \sum_i x_{ij}^2 - 2m \sum_j \sum_i x_{ij} + m^2 \sum_j n_j.$$

The value represented by this last expression is algebraically equivalent to the value of the first expression, but it may be far more convenient actually to compute this number in the second way, or to use this second expression in a mathematical argument.

A.3 SUMMATION RULES FOR TWO OR MORE VARIABLES

So far we have acted as though each distinct individual or observation were associated with one and only one numerical score. However, it may be that two distinct values x_i and y_i are paired with each individual or observation i . Then the preceding rules may be extended:

RULE 5. If each of N distinct individuals i has two scores, x_i and y_i , then

$$\sum_{i=1}^N x_i y_i = x_1 y_1 + x_2 y_2 + \cdots + x_N y_N.$$

Notice that this symbol means that the *product of the pair of scores* belonging to any given individual i is to be found *first*, and then when this has been done for the entire set of individuals, the *sum* of these products is to be found. Be sure to notice that *in general*

$$\sum_i x_i y_i \neq \left(\sum_i x_i \right) \left(\sum_i y_i \right).$$

The other rules extend quite easily to the situation where two (or more) variables are considered:

RULE 6. $\sum_i a x_i y_i = a \sum_i x_i y_i.$

RULE 7. Given the constants a and b ,

$$\sum_i (a x_i + b y_i) = a \sum_i x_i + b \sum_i y_i.$$

Sometimes it happens that when observations are arranged into groups, all observations i in the same group j will have the same value x_j on one variable, but different values y_{ij} on the other. Then, the following rule applies:

RULE 8. $\sum_{j=1}^J \sum_{i=1}^{n_j} x_j y_{ij} = \sum_j x_j \left(\sum_i^{n_j} y_{ij} \right).$

All of these rules may be extended by obvious analogy when more than three variables are involved, or when there is multiple summation.

Like most of the mechanical skills of mathematics, real facility in algebraic manipulation of summations comes only with practice. However, when puzzled or in doubt about the equivalence to two sums, you can usually check this equivalence by actually writing out the terms in the sums symbolized.

EXERCISES—RULES OF SUMMATION

Write out in extended form the sums represented by each of the following expressions:

1. $\sum_{i=1}^5 x_i$

3. $\sum_{i=1}^3 x_i^3 + \sum_{i=5}^8 x_i^2$

5. $\sum_{i=1}^3 4 \left(\sum_{j=1}^2 x_{ij} \right)$

7. $\left(\sum_{i=1}^4 x_i^2 g_i \right) / \left(\sum_{i=1}^4 g_i \right)$

9. $\sum_{i=1}^5 \sum_{j=2}^4 (x_{ij}^2 + y_i)$

2. $\sum_{i=1}^6 x_i^2$

4. $\sum_{i=2}^4 10(x_i + y_i)$

6. $\sum_{j=1}^4 \left(\sum_{i=1}^3 x_{ij} \right)^2$

8. $\sum_{j=2}^5 n_j \left(\sum_{i=1}^3 x_{ij} \right)^2$

10. $\sum_{i=4}^6 \sqrt{x_i + 10}$

Express each of the following extended sums in the appropriate summation notation:

11. $(x_1 + x_2 + x_3 + x_4)/N$

12. $x_1 y_1 + x_2 y_2 + x_3 y_3 + \cdots + x_N y_N$

13. $x_2 f(x_2) + x_3 f(x_3) + x_4 f(x_4)$

14. $p(x_1) + p(x_2) + p(x_3) + p(x_{10}) + p(x_{11}) + p(x_{12})$

15. $3(x_1 + y_1)^2 + 3(x_1 + y_2)^2 + \cdots + 3(x_1 + y_N)^2$

16. $12(x_1^2 - 2x_1 + 3) + 12(x_2^2 - 2x_2 + 3) + 12(x_3^2 - 2x_3 + 3)$

17. $[3x_1 + 3x_2 + 3x_3 + 3x_4][5x_1 + 5x_2 + 5x_3 + 5x_4]$

18. $\sqrt{(x_1 - 2)(y_1 + 5) + (x_2 - 2)(y_2 + 5) + \cdots + (x_N - 2)(y_N + 5)}$

19. $[4(x_2 - 5)^2/y_2] + [4(x_3 - 5)^2/y_3] + [4(x_4 - 5)^2/y_4]$

20. $(x_{11} - M_1)^2 + (x_{12} - M_1)^2 + (x_{13} - M_1)^2 + (x_{21} - M_2)^2 + (x_{22} - M_2)^2 + (x_{23} - M_2)^2$

21. $(x_1 + x_2 + x_3 + x_4)^2 (y_1^2 + y_2^2 + y_3^2 + y_4^2)$

Reduce each of the following expressions into simplest form by application of the rules of summation:

22. $\sum_{i=1}^{10} (5x_i - 3)$

23. $\left(\sum_{i=1}^5 (x_i + 2c) \right) \left(\sum_{i=1}^3 7 \right)$

24. $\sum_{i=4}^6 (x_i - 2a)^2 + \sum_{i=1}^3 (x_i + 2a)^2$

25. $\sum_{i=1}^N \left[x_i - \left(\sum_{i=1}^N x_i / N \right) \right]$

$$26. \sum_{i=1}^N (x_i - 3)(x_i + 3)/N$$

$$27. \sqrt{\sum_{i=1}^N (a+b)} \sqrt{\sum_{i=1}^N c}$$

$$28. \left[\left(\sum_{i=1}^N x_i \right)^2 - \left(\sum_{i=1}^N y_i \right)^2 \right] / \sum_{i=1}^N (x_i + y_i)$$

$$29. \frac{\sum_{i=1}^N \left[x_i^2 - \left(\sum_{i=1}^N x_i/N \right)^2 \right]}{N}$$

$$30. (1/N) \sum_{i=1}^N \left[x_i - \left(\sum_{i=1}^N x_i/N \right) \right] \left[y_i - \left(\sum_{i=1}^N y_i/N \right) \right]$$

Consider the following values:

$$x_1 = 7 \quad x_4 = 4 \quad x_7 = 2 \quad x_{10} = 5$$

$$x_2 = 3 \quad x_5 = 0 \quad x_8 = 8$$

$$x_3 = 9 \quad x_6 = 1 \quad x_9 = 8$$

In terms of these values for x_i , evaluate the following sums:

$$31. \sum_{i=1}^{10} (x_i - 5)$$

$$32. \sum_{i=1}^{10} (x_i^2 - 3x_i + 1)$$

$$33. \sum_{i=1}^5 x_i - \sum_{i=6}^{10} (x_i + 5)$$

$$34. \sum_{i=1}^{10} \left[x_i - \left(\sum_{i=1}^{10} x_i/10 \right) \right] / 10$$

$$35. \left(\sum_{i=1}^{10} x_i \right) / \left(\sum_{i=1}^5 x_i \right)$$

$$36. \left(\sum_{i=1}^{10} x_i^2/10 \right) - \left(\sum_{i=1}^{10} x_i/10 \right)^2$$

In terms of these values for x_i , evaluate the following sums:

$$37. \sum_{i=1}^{10} (x_i - 5)$$

$$38. \sum_{i=1}^{10} (x_i^2 - 3x_i + 1)$$